

Final Project (Total: 100 points)

RISC-V Multi-cycle Processor

Design RISC-V Multi-cycle Microprocessor. Test sequence is provided by file (riscvtest.txt). It contains series of 32bit binary codes. “riscvtest.s” file shows equivalent assembly code of each binary codes for your reference. You can expect how controller (FSM), Datapath and Data Memory should work during simulation. Please copy top test bench (top_testbench.v) and those test sequence files from “/afs/ee.cooper.edu/dist/koo/ECE447”.

Data/Instruction Memory is defined at “instr_data_mem.v” file so please design Controller and Datapath. Refer pages “Multicycle Processor Main FSM” and “Multicycle RISC-V Processor” in class materials. Fig.1 and Fig.2 shows snapshot of these slides.

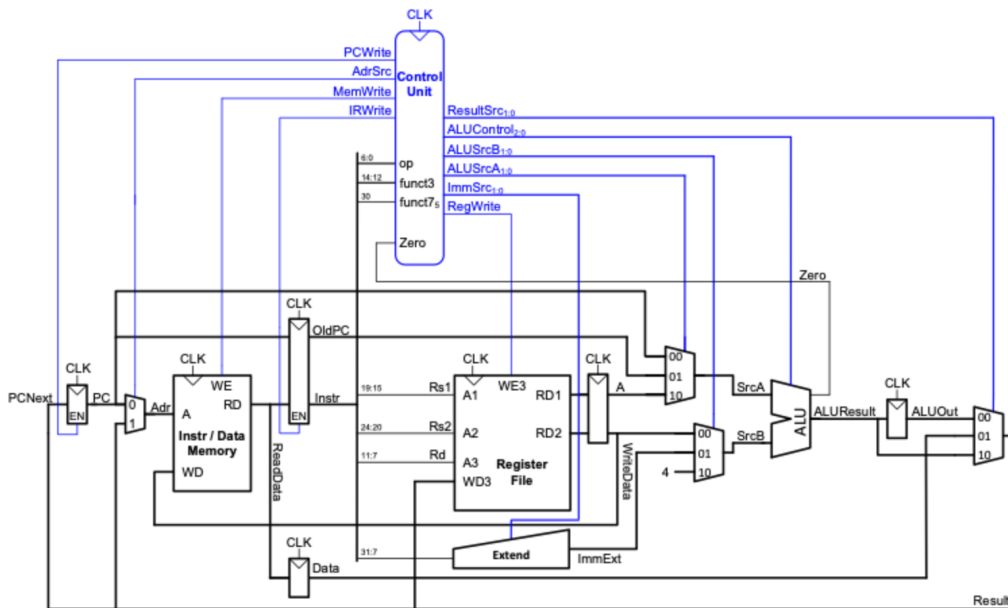


Fig.1 Block diagram of RISC-V Multicycle Processor

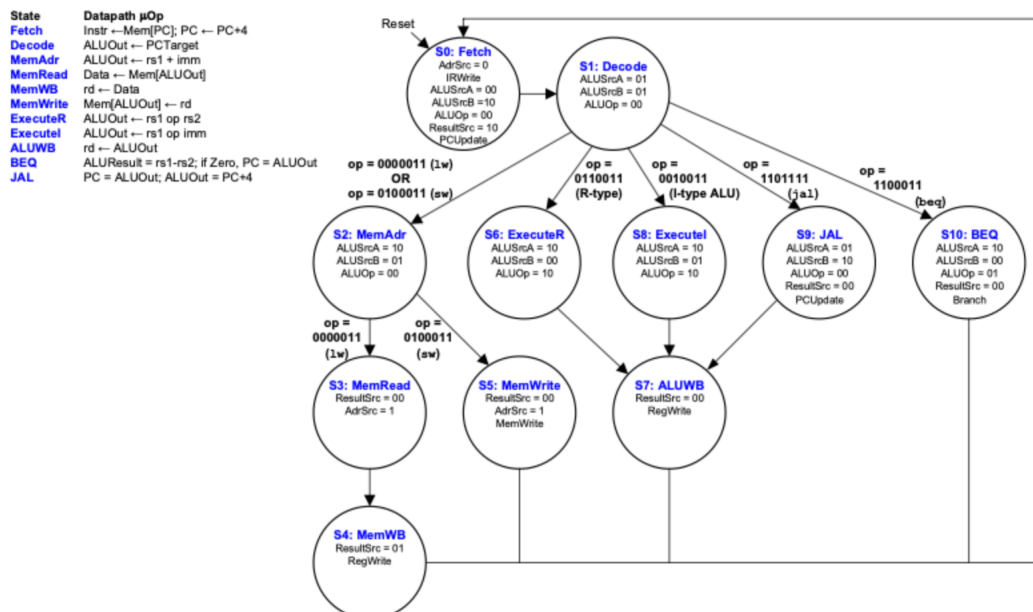


Fig.2 State diagram for Main FSM

In Fig.1, please keep in mind that net “Result” and “PCNext” is identical net. For grading points, please refer below statements.

- Verify your Verilog simulation works. As shown in Fig.3, you need to show how each status of FSM flows to process each assembly command, and include snapshot like Fig.3 in the final report.

For example, after reset, the first command “addi” is processed by FETCH (0)->DECODE (1)->EXECUTEI (8)-> ALUWB (7) flow. For easy follow and debug, please define same number for each status in FSM as shown at Fig.2.

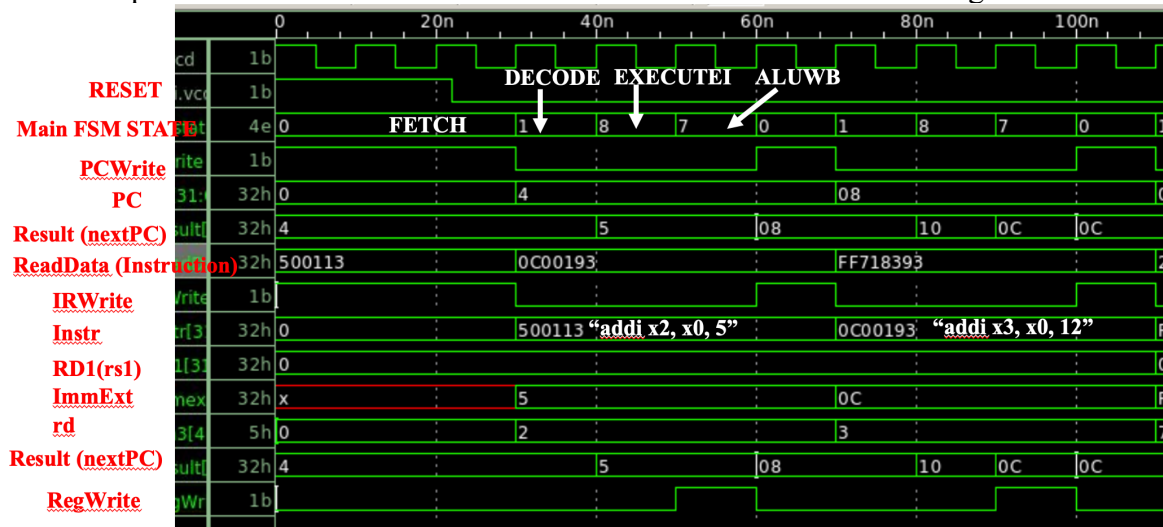


Fig.3 Simulation results for processing very first two commands “addi”

- When test bench completes with the message “Simulation succeeded” : **80 points**
- If top simulation does not work (partial points):
 - The first 3 “addi” command works: 15 points
 - Next “or” command works: 2 points
 - Next “and” command works: 2 points
 - Next “add” command works: 8 points
 - Next “beq” command works: 4 points
 - Next “slt” command works: 4 points
 - Next “beq” command works so jump to “around”: 4 points
 - “sub” command works: 2 points
 - “sw” command works: 4 points
 - “lw” command works: 2 points
 - “Jal” command works: 2 points
 - None of command works: 1 points
 - Complete Data path with the correct connections for each modules: 30 points
- Synthesis flow succeeds with positive number for Slack time in *.min.fullpath.rpt report file: **20 points**.
 - When you synthesize, exclude ‘testbench’ and ‘mem’ modules. Try to synthesize ‘controller’ and ‘datapath’ modules.
- If synthesis succeeds with 1GHz clock period, **20 bonus points** will be added.
- Modules for Register files and extend units are defined in the top testbench (top_testbench.v)